TECHNICAL ASSESSMENT

ROV Telemetry Ground Station

Build one native desktop application that simulates a small underwater vehicle and lets an operator monitor and control it.

Application requirements

- Simulate movement forward, sideways, up/down, and turning, including simple drag and a short thruster-response delay.
- Display live armed state, mode, roll, pitch, yaw, depth, and battery voltage.
- Provide Arm/Disarm controls and Manual, Depth Hold, and Heading Hold modes.
- In Manual mode, accept keyboard or joystick controls.
- In Depth Hold and Heading Hold modes, let the operator set a target depth or heading; the holding method is your choice.
- Safety: thrust must be zero whenever disarmed. In Manual mode, return thrust to zero after one second without operator input.
- Display a live depth-versus-time plot and an OpenStreetMap view inside the application.
- Keep the application responsive during simulation and interface updates.

Video walkthrough application display only

- Show the main desktop application window and dashboard layout.
- Display sample or simulated values for armed state, mode, roll, pitch, yaw, depth, and battery voltage.
- Show the Arm/Disarm controls, Manual / Depth Hold / Heading Hold modes, and target depth/heading inputs.
- Show the depth-versus-time plot with sample or simulated data and the OpenStreetMap view.
- The recording is an interface walkthrough only. Working movement, hold modes, and safety responses do not need to be demonstrated in the video; be ready to explain your implementation at interview.

Rules and technical guidance

- C++ with Qt/QML is preferred. Python or another suitable desktop language is acceptable. Do not build a browser or web application.
- No hardware is required. Simple simulated or sample data (e.g., depth, heading, orientation, battery voltage) may be generated directly in the application. SITL, MAVLink, ROS, or similar frameworks are optional and give no extra advantage.
- AI coding tools are permitted; disclose which were used and how in the README.
- You must be able to explain and modify every part of the implementation during the interview.
- Prefer a smaller, reliable implementation over an incomplete feature set.
- Optional extras: CSV logging, simulated water current, or unit tests.